

CS336 Assignment 3: Writeup

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2.1 chinchilla_isoflops

- (a) Show your extrapolated compute-optimal model size, together with the $\langle C_i, N_{\text{opt}}(C_i) \rangle$ points you obtained. What is your predicted optimal model size for a budget of 10^{23} FLOPs? What about for 10^{24} FLOPs?

Answer: The red points are the observed $\langle C_i, N_{\text{opt}}(C_i) \rangle$ points, and the blue line is the fitted scaling law. With a budget of 10^{23} FLOPs, the predicted optimal model size is 5.00×10^{10} parameters. With 10^{24} FLOPs, the predicted optimal model size is 1.27×10^{11} parameters.

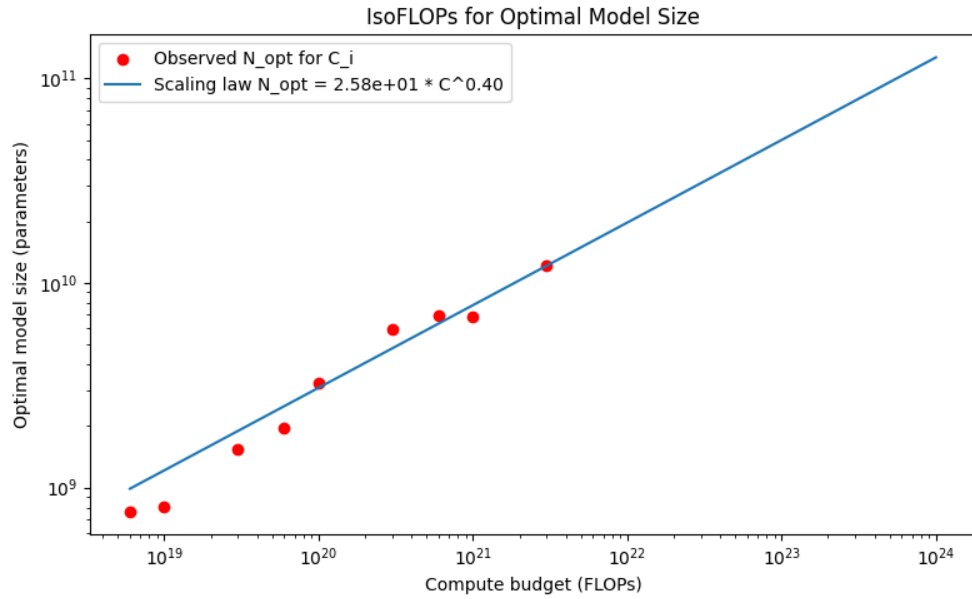


Figure 1: Extrapolated compute-optimal model size

- (b) Show your extrapolated compute-optimal dataset size, together with the $\langle C_i, D_{\text{opt}}(C_i) \rangle$ data points from the training runs. What is your predicted optimal dataset size for budgets of 10^{23} and 10^{24} FLOPs?

Answer: The blue points are the observed $\langle C_i, D_{\text{opt}}(C_i) \rangle$ points, and the black line is the fitted scaling law. With a budget of 10^{23} FLOPs, the predicted optimal dataset size is 3.37×10^{11} tokens. With 10^{24} FLOPs, the predicted optimal dataset size is 1.33×10^{12} tokens.

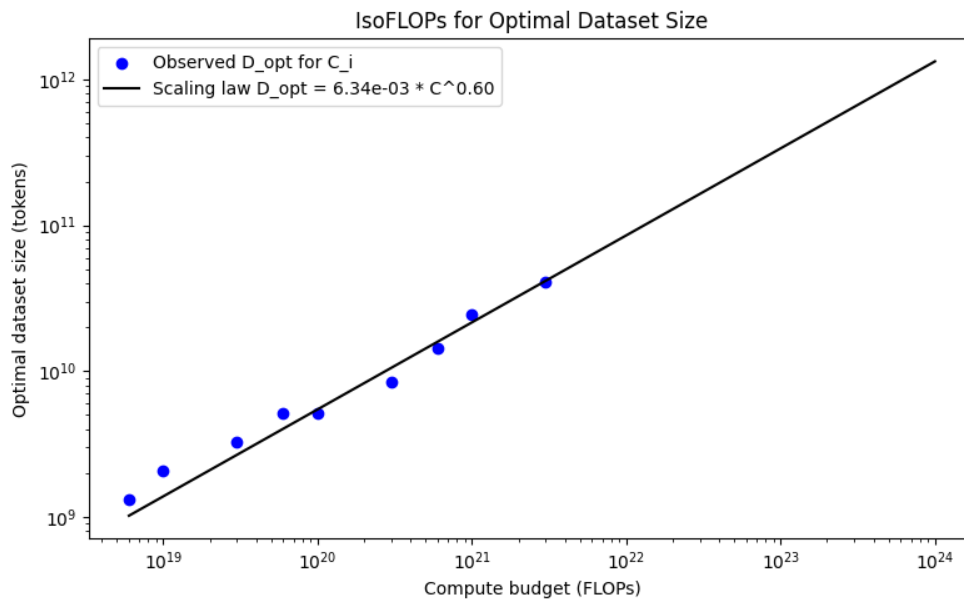


Figure 2: Extrapolated compute-optimal dataset size

3 scaling_laws

Answer: